



[GCT555 3DID]

ProxiWall: Seamless Private-to-Shared Mode Switching for Collaborative 3D Exploration on Dual-Wall Displays

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Content

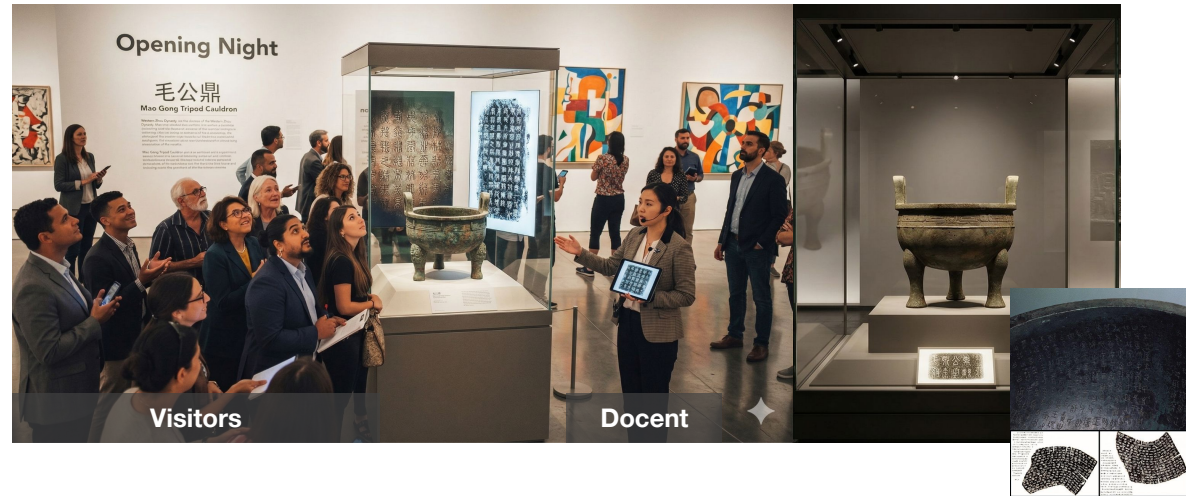
1. Main Concept
2. Interview #1
3. Interview #2
4. Project focus



Main Concept-Initially

Dual wall collaborative 3D exploration for museums

- Shared viewing
- Private exploration
- Distance based transition





Main Concept-Now

Dual wall collaborative 3D exploration for museums

- Distance based transition
- Gesture based 3D object manipulation
 - Geometric manipulation
 - Semantic manipulation



Interview #1 - Dooyoung Kim, Ph.D

Questions We Asked

- Is this museum scenario meaningful?
- Are user roles clear?
- Does shared/private exploration make sense?
- What use case is strongest?

Key Feedback

- Role distinction improves clarity
- Curator ↔ Visitor is compelling
- 3D artifact viewing has clear value
- Educational / cultural context is strong



Interview #2 - Woojin Cho, Ph.D

Questions We Asked

- How should users manipulate the object?
- Is switching enough as contribution?
- What concrete interactions are needed?

Key Feedback

- Need stronger interaction contribution
- Need visible input-feedback mapping
- Need manipulation beyond movement
- Need to scope problem and environment definition



What Is New?

A dual-wall museum interaction system that combines **proxemic** mode switching + **multimodal** 3D manipulation + **shared/private merge** mechanics



Modified System

- **Environment setting**

- a. **Input** Multiple RGB-D Camera (Front view + Top view)
- b. **Output** 3D Rendered Video — Both-side LED Wall (w/o HMD)

- **Interaction feature**

- a. Distance-based sharing mode transition
- b. Parts selection pseudo touch
- c. Gesture-based geometric manipulation
- d. Sound-based merge or discard



Modified System

- **Interaction feature**
 - (a) Distance-based sharing mode transition

Initial design : mode switching

Private/Share mode switching was positioned as the core contribution of the project.

Automatic transition triggered by user's distance from the LED wall — stepping back enters Share Zone, stepping forward enters Private Zone.

Revised design

- 1) Zone boundaries visualized on the floor as semicircular regions to augment spatial awareness
- 2) Project reframed as "wall-oriented shared virtual object collaboration interface"
- 3) Mode transition now serves as contextual state for other interaction features



Modified System

- **Interaction feature**
 - (b) Parts selection pseudo touch

Initial design: object selection

Artifacts treated as single monolithic objects.

Only a simple scenario of choosing one artifact among several existed.

No mechanism for selecting or manipulating individual components within an artifact.

Revised design : [Pseudo]

- 1) Introduced **multi-part artifact decomposition structure**.
- 2) Segmentation-based **UI panel recognition + fingertip-to-screen contact detection** for pseudo touch
- 3) Users select individual parts, manipulate them independently, and reassemble to original position



Modified System

- **Interaction feature**
 - (c) Gesture base geometric manipulation

Initial design

Only an abstract description of "manipulating 3D objects" with no concrete input method defined.

Exploratory ideas such as mirror-based touch simulation on laptop screens were discussed but not finalized.

Revised design

- 1) Defined a concrete gesture pipeline **using ceiling-mounted cameras + MediaPipe**.
- 2) Hand orientation and position directly linked to virtual object transformation
- 3) **Counter-directional rotation**: object rotates opposite to user movement for efficient multi-angle exploration



Modified System

- **Interaction feature**
 - (d) Sound-based merge or discard

Initial design

No mechanism for propagating Private-mode modifications to the Share view.

Mode transition existed, but object state behavior upon switching was undefined.

Revised design

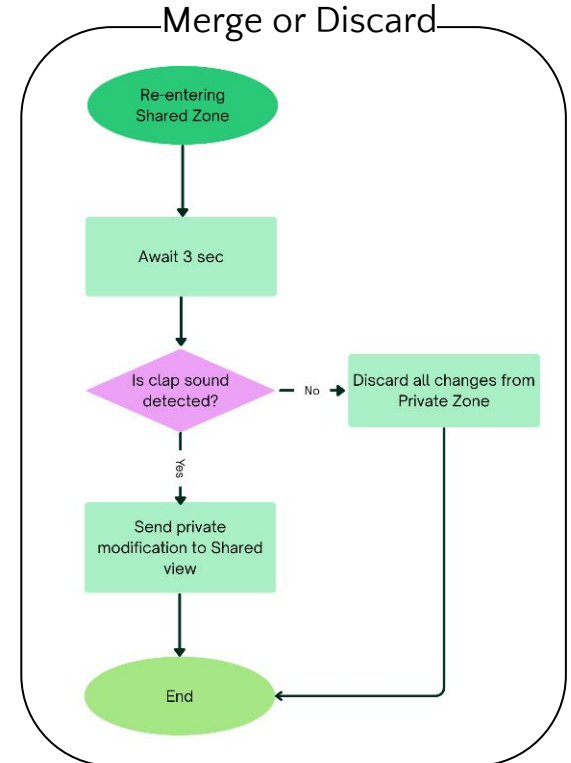
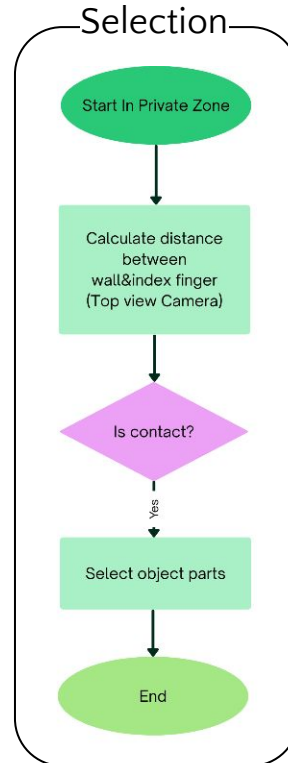
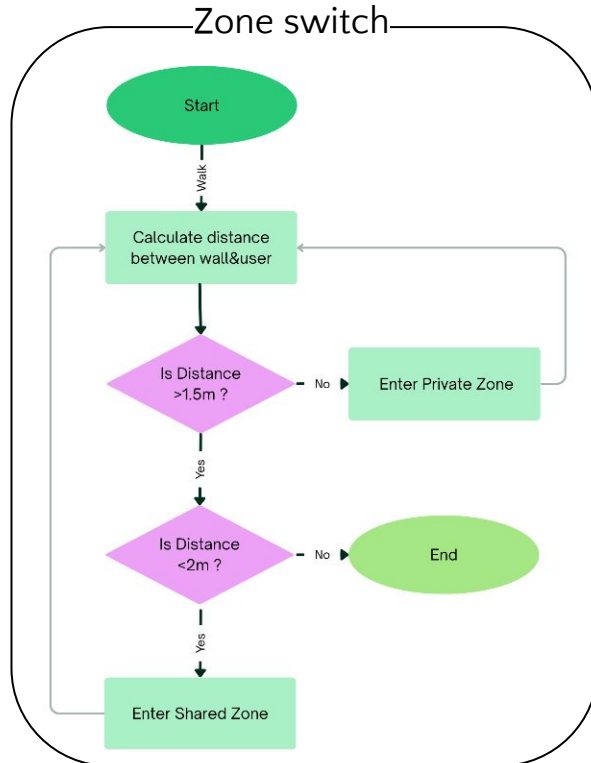
1) Introduced multimodal input (camera + sound) for intentional commit/rollback.

- Exp) Clap within 3 sec → modifications merge into Share view (magnetic stitching effect)
- Exp) No clap → object reverts to original state (brush-out dismissal effect)

2) Transforms simple mode switching into a deliberate user-driven decision mechanism

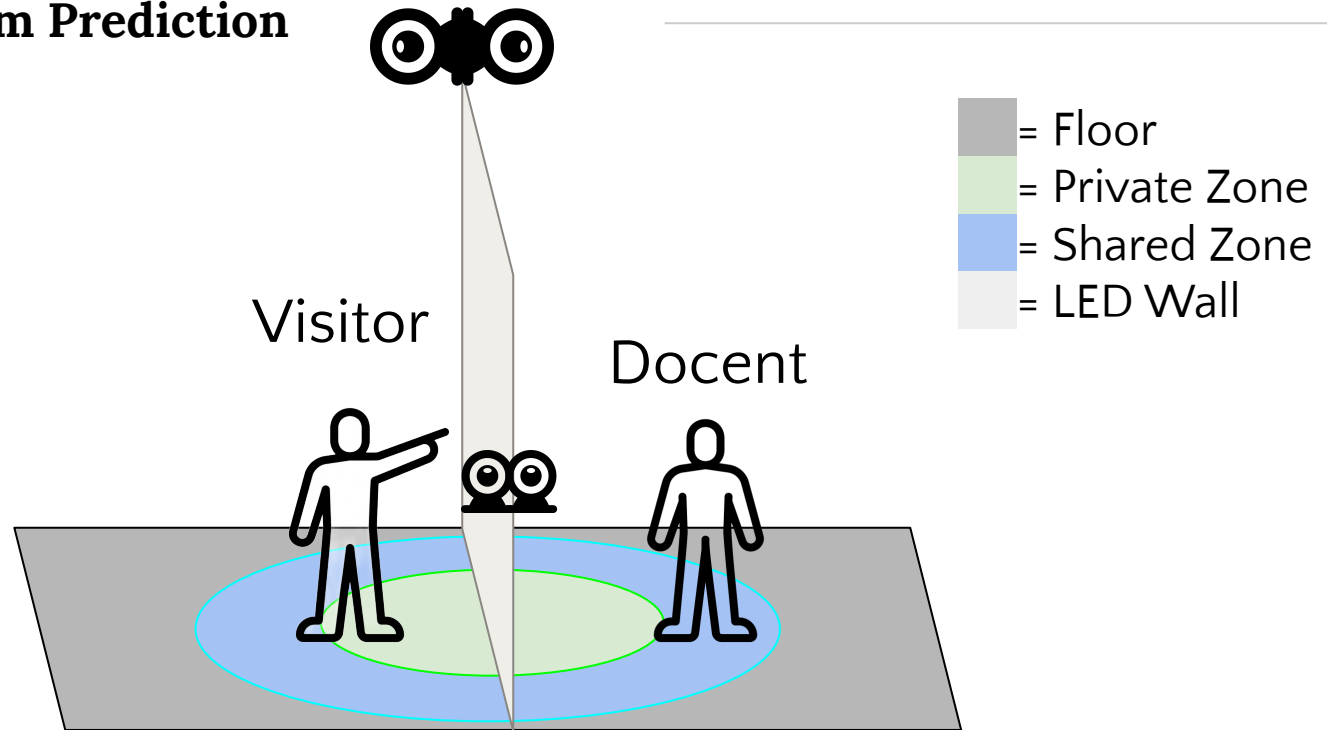


Interaction System Flow



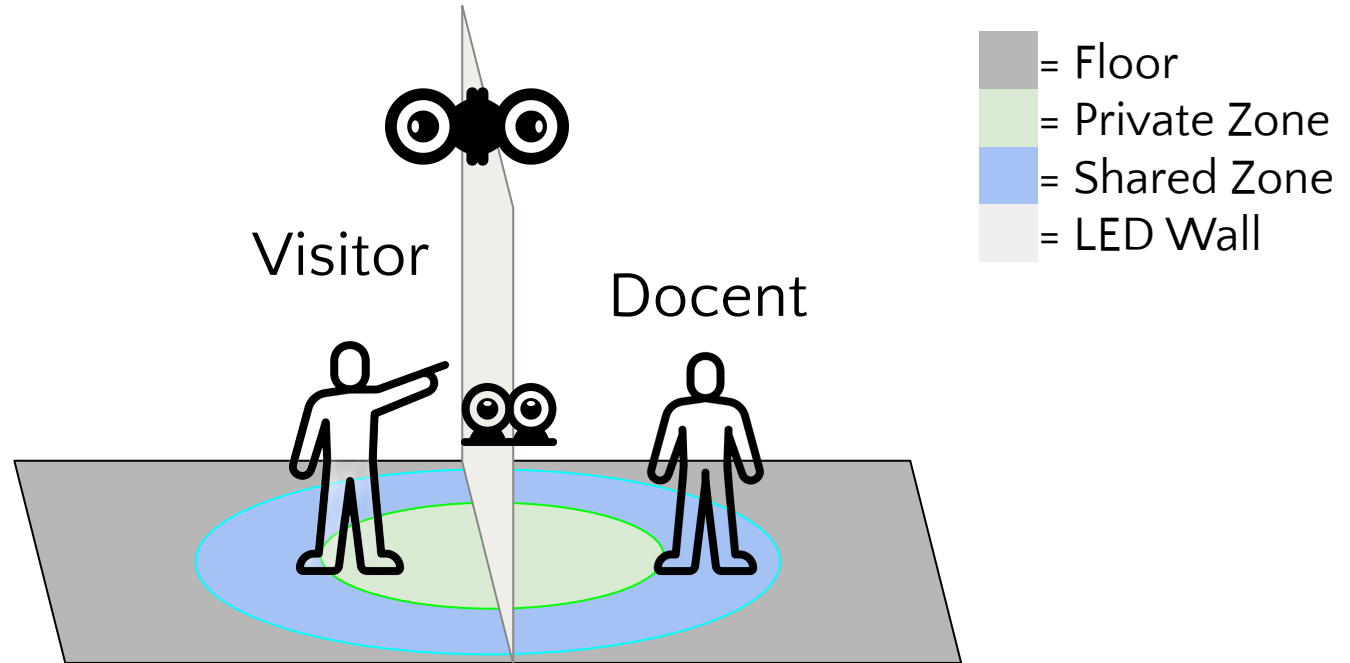


System Prediction





System Prediction



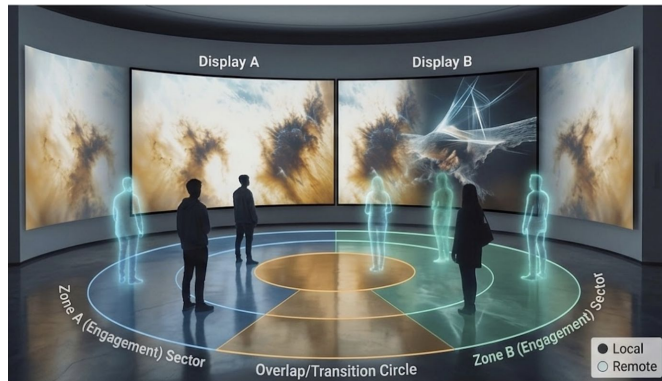


Core Contribution

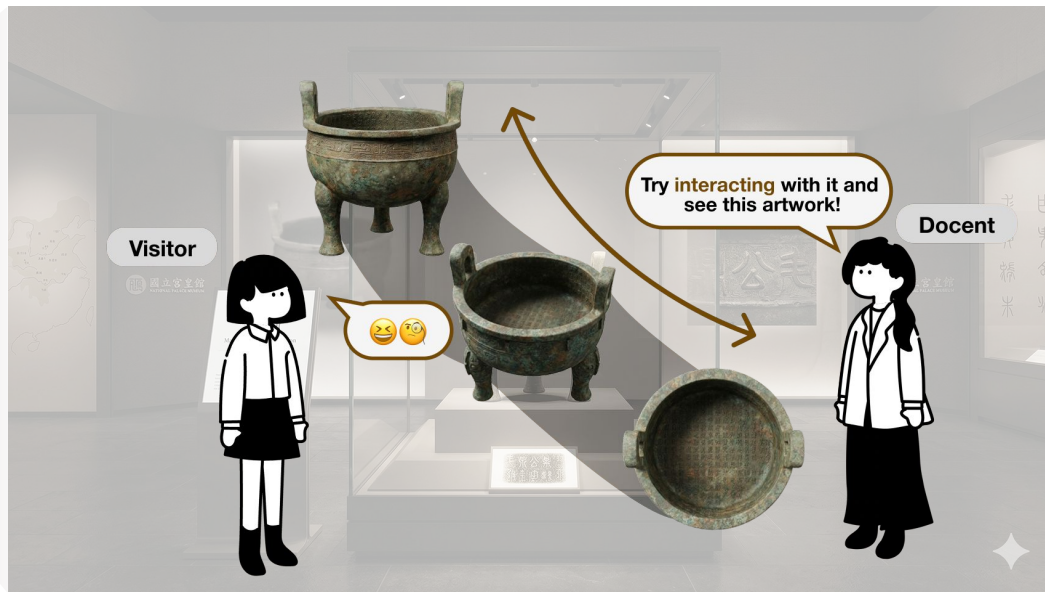
- **Validated museum collaboration scenario**
- **Seamless proxemic zone transition**
- **Rich multimodal 3D manipulation**
- **Shared/private collaboration workflow**



Q&A



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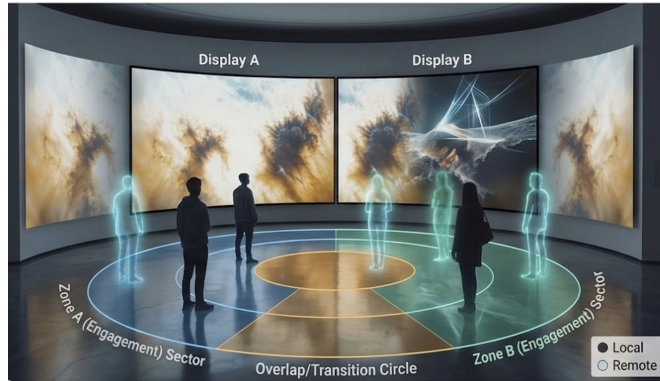


Needs (Target User)

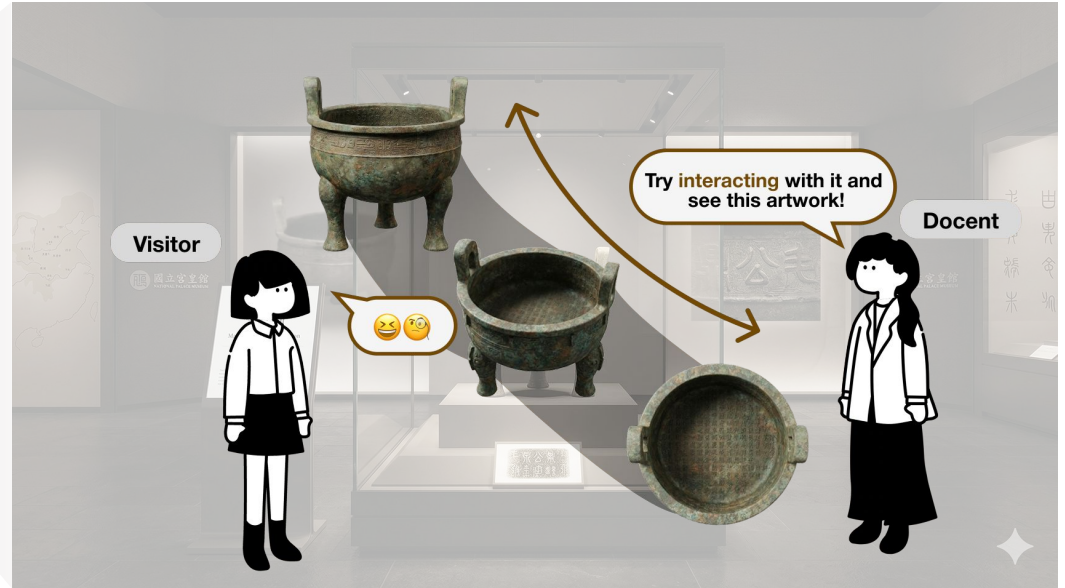




Approach



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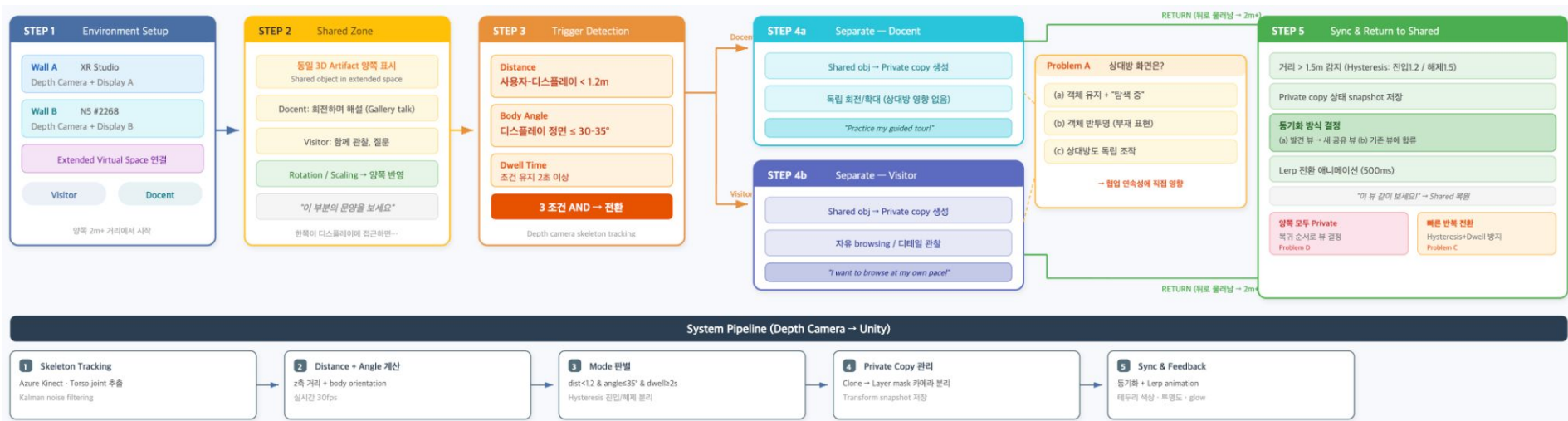


wall displays with spatial zones

each user's viewpoint angle to create a shared space



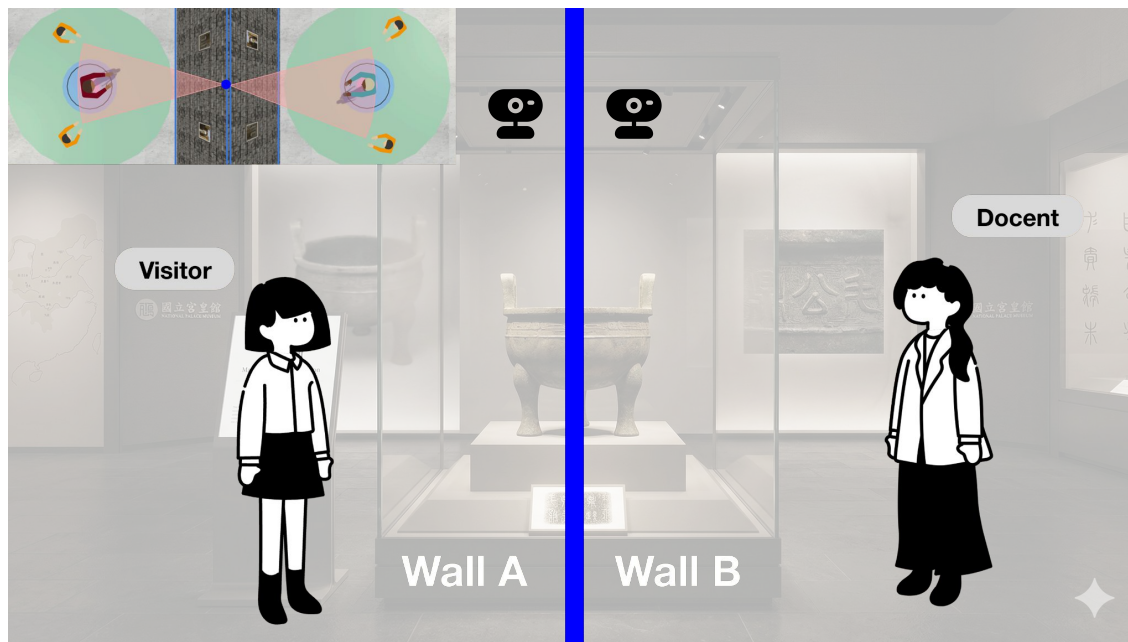
UX Scenario (Flow Chart)





UX Scenario

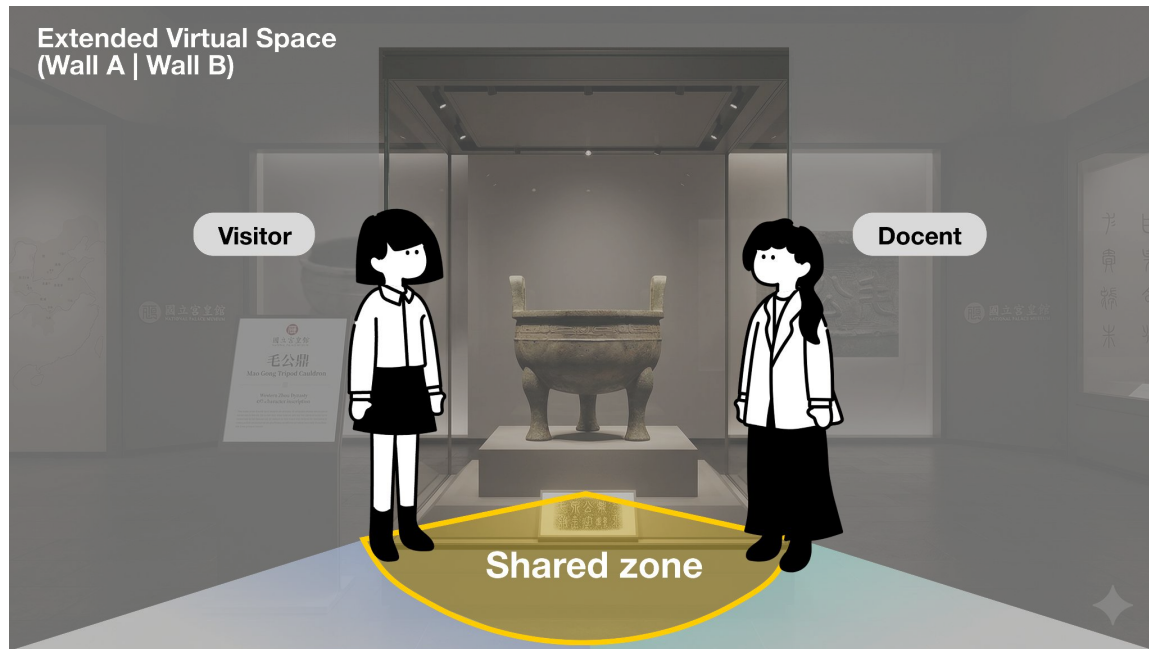
(1) Environment Setup : 2 Camera, Wall Display A (XR Studio), B (N5 #2268)





UX Scenario

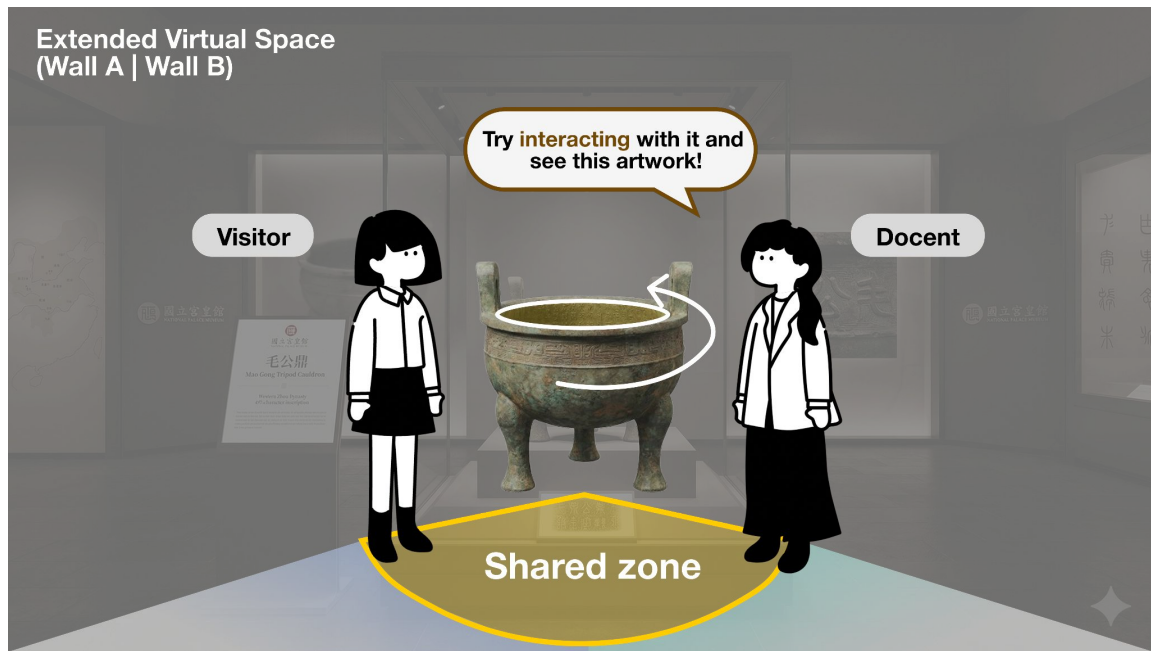
(2) Extended Virtual Space (Wall A | Wall B) -> Shared Zone : Walk into intersection zone.





UX Scenario

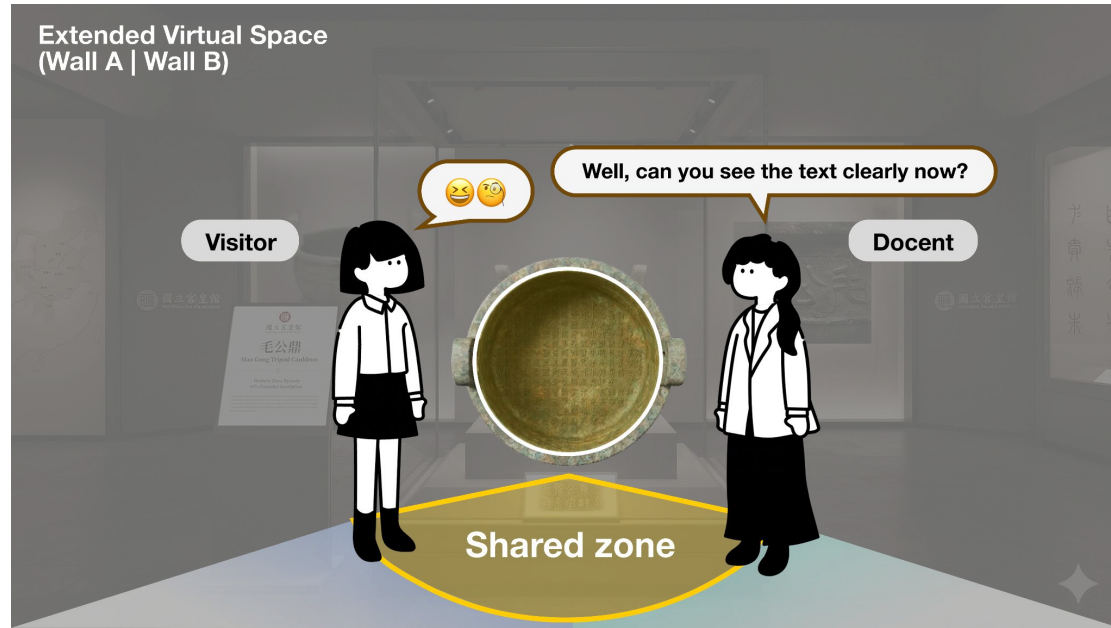
(3) Shared Zone : Seamlessly transition to 3D artwork exploration interaction without any explicit input





UX Scenario

(3) Shared Zone : 3D Artwork Exploration Interaction (Object Rotation, Scaling)



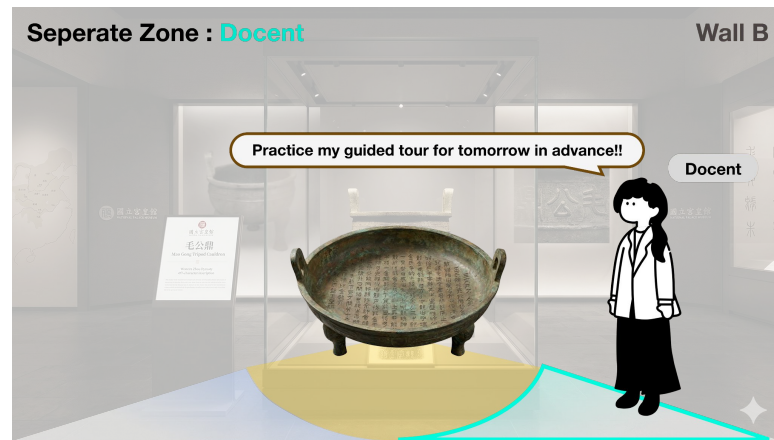


UX Scenario

(4) Separate Zone

Visitor : When moving beyond the viewpoint angle, the visitor (user) enters a separate zone.

Docent: Docent with collaborative interaction for guided tour.

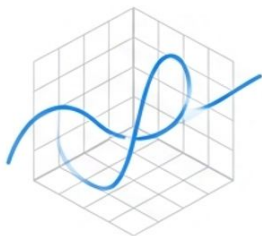




Benefit

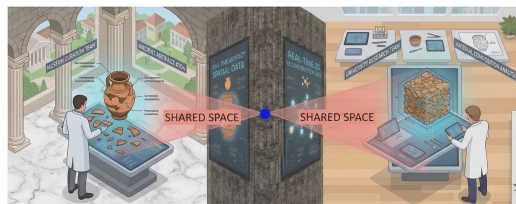
Expected results and 3 key contributions

Fluid Transition



Using natural body movement, we can examine the objects privately during an ongoing docent tour.

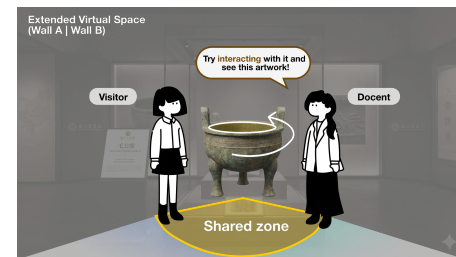
Dual-Wall Collaboration



Wall Display (A) Wall Display (B)

A wall with a depth camera enables smooth collaboration between docent and visitor.

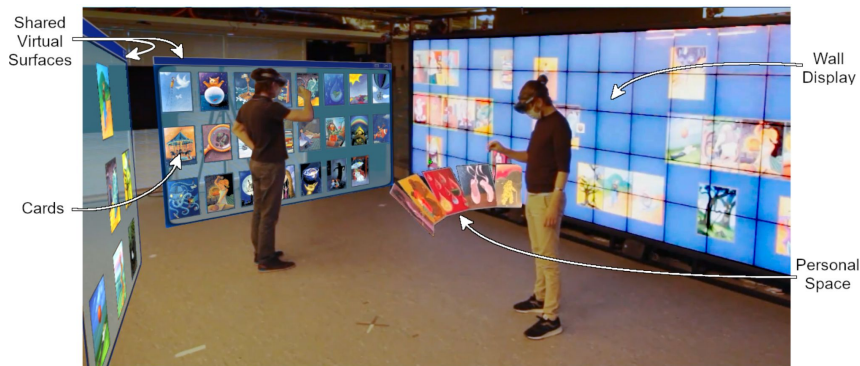
Empirical Design Guidelines



Empirical evidence from a user study on distance threshold, visual feedback strategies, and the impact of private exploration



(1) Comparison : Wall Displays + AR



Evaluating the Extension of Wall Displays with AR for Collaborative Work.

James, R., Bezerianos, A., & Chapuis, O. (2023, CHI).

Proposed Workspace System: Wall Displays + AR

Pros

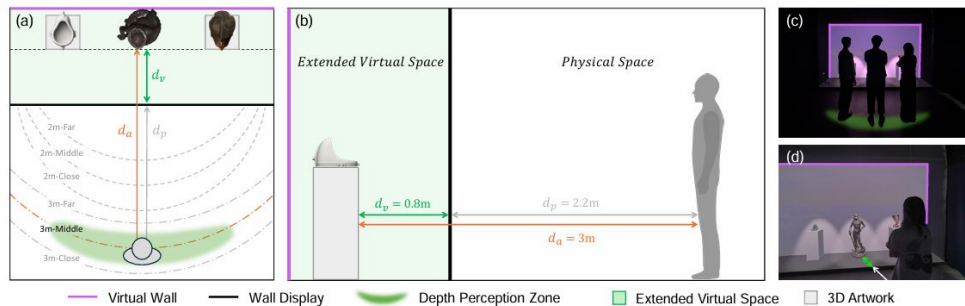
- Evaluated the combination of Wall Displays + AR against standalone Wall Display usage
- Wall Displays serve as Co-located Space, while AR functions as Personal Space
- Most participants preferred the Wall+AR condition — rated as more enjoyable and immersive

Cons

- Limited to collaborative scenarios where Wall Displays are used as a shared workspace
- No significant improvement in task performance
- Extends the workspace conceptually, not the user's physical space



(2) Comparison : Intersection zone using angle view point



Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display

in IEEE Transactions on Visualization and Computer Graphics (ISMAR-TVCG / 2025)

Pros

- **Autonomy for Private Exploration:**
By simply stepping into the private zone, a user can independently manipulate the 3D object.
- **Systematic Management of Workspace Awareness:**
Controls the visual feedback during private mode to maintain a seamless flow of collaboration.

Cons

- **Potential Disruption of Co-presence:**
The private mode may negatively impact their sense of togetherness and engagement.



(3) **Comparison** : Shared space telepresence system



Project Starline: A high-fidelity telepresence system

in ACM SIGGRAPH 2024 Emerging Technologies, pp.1–2, 2024.1

Pros

Larger Space: The system provides a larger area where users can freely walk around and interact intuitively based on their physical distance.

Divided Private/Shared Space: The zones are clearly separated by the user's distance from the display. This allows users to control what they want to share with their partner and what they want to view independently.

Cons

Immersion: The overall immersion and feeling of co-presence might be lower compared to systems like Starline, which provide a highly realistic illusion of a real person.



Deliverables

An interactive system for dual-wall display environments that leverages the user's physical distance as an implicit input to naturally transition between private and shared exploration modes.

